Jackson Hellmers

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Education

Master of Science in Electrical Engineering GPA: 4.00/4.00
University of Wisconsin-Madison Sep 2021 - Sep 2022

Bachelor of Science in Electrical and Computer Engineering GPA: 3.95/4.00
University of Wisconsin-Madison Sep 2017 - May 2021

Highlighted Coursework: Graduate Machine Learning, Artificial Intelligence, Neural Networks, Image Processing, Computer Vision, Probabilistic Modeling, Signal Processing

Programming Skills

Languages: Python, Java, C/C++, MATLAB, SQL
Machine Learning Libraries: Tensorflow, PyTorch, Keras, Scikit-learn, Numpy, Pandas, Matplotlib, OpenCV

Work Experience

Machine Learning Engineer, Milwaukee Electric Tool Company, *Brookfield, WI* June 2022 - Current

- Developed and deployed embedded machine learning pipelines for real-time tool-state and event detection using high-rate (1 kHz) IMU telemetry.
- Built production C/C++ inference paths from Python-trained models, including export of model parameters for on-device evaluation and decision logic.
- Engineered and optimized signal-processing feature extraction (statistical, derivative, and low-frequency spectral features) for constrained firmware targets.
- Created Python-to-firmware parity test harnesses to validate end-to-end prediction consistency and reduce regression risk during model and firmware updates.

Software Engineering Intern, Innovative Signal Analysis, *Richardson, TX* May 2021 - Aug 2021

- Programmed Xilinx UltraScale RFSoC boards using Vivado Design Suite and Vitis IDE.
- Generated bare metal applications to benchmark Real-time Computing cores.
- Designed Verilog modules to measure latency between a system's CPU and FPGA.
- Created PCBs to measure the power draw of various components using fixed-voltage supply and current sense ICs.

Design Verification Test Engineer, Extreme Engineering Solutions, *Madison, WI* Jan 2020 - Sep 2020

- Worked with high-bandwidth (50 GHz) oscilloscopes to verify clock and data signal functionality
- Constructed and improved PCB designs to adjust clock slew rates and power supply switching speeds.
- Recorded daily activities and software adjustments through online tickets and version control applications.
- Documented verification tests measuring the compliance of numerous I/O interfaces such as USB 3.0, 100 Gig Ethernet and PCIe 4.0.

Systems Development Intern, Exact Sciences, *Madison, WI* May 2019 - Dec 2019

- Designed and prototyped simple DC motor control and voltage regulation circuits.
- Developed and ran autonomous data-driven C++ programs on Arduino boards and Python programs on Raspberry Pi boards.
- Followed wiring diagrams and guidelines to assemble industrial sized automated systems.
- Used regression models to determine product quality using a variety of sensor data.

Projects

DeViSE - Deep Visual Semantic Embedding, Academic Project

- Inspired by and adapted from **Google Research Paper:** DeVise - A Deep Visual-Semantic Model
- Combined a trained visual model with an embedded semantic model to allow for classification of unsupervised image labels by taking advantage of contextual similarity.

Image Super-Resolution, Academic Project

- Compared State-of-the-Art CNN and GAN models to evaluate the advantages of each model.
- Created CNN that outperforms general upscaling methods such as bicubic interpolation and Gaussian denoising.